

Technical Proof of Concept Plan: Scalable Multi-Tenant LLM Microservice

IDEA_INFORMATION: Develop a **scalable, multi-tenant microservice** for **real-time document summarization** using a **Fine-Tuned LLM**, processing **100 documents/sec** with **95% summarization accuracy** (measured against a human-validated corpus).

1. PoC Hypothesis and Success Criteria (SMART)

PoC Hypothesis: The combined **Mistral/Qwen MoE model cascade**, when containerized via **Docker** and served by **FastAPI**, can achieve **95% summarization accuracy** on the target document corpus while sustaining a **p95 latency** of **< 200ms** per document inference under a concurrent load simulating **100 documents/sec**.

KPI	Metric	Target	Rationale
Accuracy	F1 Score / ROUGE-L	≥ 0.95	Validates technical feasibility and business value of the fine-tuned model.
Performance	p95 Inference Latency	≤ 200 ms	Validates the "real-time" requirement under high load for a quality user experience.
Throughput	Sustained RPS (Requests/sec)	≥ 100 RPS	Validates architectural scalability and resource efficiency under required load.
Efficiency	Container Image Size (Inference)	≤ 5 GB	Demonstrates production-intent optimization for faster deployment and lower operational cost.
Tenancy Isolation	Request correlation and data separation	100% (Binary)	Validates that logs and data for concurrent, simulated tenants are strictly isolated.

2. Architectural Blueprint (High-Level)

The architecture is minimal, container-native, and optimized for MLOps validation.

Component	Technology	Rationale (FOSS Focus)
API Gateway	Nginx (or Traefik in a K3s setup)	Lightweight, high-performance reverse proxy for load balancing and tenant-specific routing.
Core Service	Python (FastAPI) + Uvicorn	Asynchronous (async/await) capability for high concurrency

Component	Technology	Rationale (FOSS Focus)
		(I/O and CPU-bound tasks), fast API development.
Inference Engine	HuggingFace Transformers, vLLM	Optimized library for accelerated LLM inference, handling quantization and MoE models efficiently.
Knowledge Retrieval (RAG)	Qdrant (Vector DB) / pgvector	Persistent, open-source vector store for multi-tenant knowledge base/document embeddings.
Deployment	Docker & Docker Compose (PoC) / K3s (Minimal K8s)	Provides containerization, environment parity, and the path to scalable orchestration.

3. Recommended Free Open-Source (FOSS) Tech Stack

3.1 AI Model Cascade

Role	Model (FOSS)	Parameter Size	Rationale
Baseline/Research Coverage	Qwen3-30B-A3B-Instruct-2507 (MoE)	~30.5 B total (3.3 B active)	Excellent long-context handling and strong comprehension. Used for initial fine-tuning and upper-bound accuracy.
Production/Speed Focus	Mistral-8x22B (SMoE)	~141 B total (39 B active)	High performance/speed for its size; competitive accuracy. Target model for achieving the ≤200 ms latency KPI.
Embedding	BGE-M3 or E5-mistral-7b	N/A (small)	High-quality, multilingually capable embedding model for RAG and vector indexing.

3.2 Core FOSS Stack

Component	FOSS Tool	Purpose
Vector Database	Qdrant (or Weaviate)	Cloud-native, high-performance vector search engine for RAG context retrieval.
Logging/Tracing	SigNoz (Unified OTLP)	Single-pane observability (Metrics, Logs, Traces) built on OpenTelemetry—superior to fragmented Prometheus/ELK

Component	FOSS Tool	Purpose
		stack for microservices.
Caching/Messaging	Redis	High-speed cache for pre-computed embeddings and rate-limiting/session management.
Orchestration (Minimal)	K3s (Lightweight Kubernetes)	Provides a quick path to a real distributed environment for high-RPS stress testing.

4. 15-Day Aggressive Timeline & Task Breakdown

Sprint Goal: Validate $\geq 95\%$ accuracy and ≥ 100 RPS performance on the core summarization microservice with robust multi-tenancy.

Day	Lead	Primary Task	Deliverable/Code Artifact
1-3	Will	Data Pipeline & Initial Training - Preprocess corpus (data cleaning, chunking), generate human-validated summary pairs. Baseline LORA fine-tuning of Qwen3-MoE .	Data pipeline scripts, Qwen3 model weights (.safetensors), Baseline Accuracy Report (F1/ROUGE-L).
4-6	Eric	Service Infrastructure & Multi-Tenancy - Setup FastAPI service with /summarize endpoint. Implement request-scoped Tenant ID extraction/validation and basic rate-limiting via Redis .	Dockerized FastAPI service, Tenant ID middleware, Redis config via Docker Compose.
7-9	Mike	Inference & RAG Integration - Integrate vLLM for optimized inference. Set up Qdrant via Docker, implement RAG lookup for context (simulate multi-tenant knowledge retrieval).	RAG embedding pipeline, Qdrant Docker setup, functional /summarize endpoint with RAG context.
10-11	Will	Model Optimization & Swap - Fine-tune Mistral-8x22B (target model) using LORA/QLORA.	Optimized Mistral weights, Model Swap documentation, Preliminary Performance Check

Day	Lead	Primary Task	Deliverable/Code Artifact
		Optimize model quantization (e.g., Q4_K) and swap inference engine from Qwen to Mistral.	(Latency/RPS).
12-13	Eric	Observability & Load Testing - Deploy SigNoz stack. Instrument service with OpenTelemetry traces (logging Tenant ID in traces). Execute structured load test (e.g., Locust) simulating 100 RPS across 10 concurrent tenants .	Fully instrumented service, SigNoz dashboard with p95 latency visualization, Load Test Report.
14-15	Mike	PoC Review & Artifact Finalization - Review performance against all KPIs. Refactor Dockerfiles for minimal size. Finalize Architectural Decision Records (ADRs) and technical documentation.	Final PoC Status Report, Cleaned/Minimized Docker images (Target 5 GB), Project Readme with clear deployment steps.

5. Technical Best Practices & Advanced Concepts (CTO Tips)

Technical Best Practices (For PoC)

- **1. Leverage Docker Multi-Stage Builds:** Strictly minimize the final Docker image size by separating the build environment (e.g., LORA tuning, large dependencies) from the final runtime environment (e.g., only vLLM and the quantized model). This is crucial for the 5 GB KPI.
- **2. Structured Logging with OpenTelemetry:** Enforce **JSON structured logging** via a lightweight collector like **Fluent Bit** to route logs to **SigNoz**. Mandate the inclusion of a unique correlation_id and the tenant_id in every log record and distributed trace to ensure 100% tenancy isolation validation.
- **3. VS Code Remote Containers:** Utilize VS Code's **Remote - Containers** extension for all development. This ensures a uniform, pre-configured environment for all three developers, eliminating "it works on my machine" issues and guaranteeing environment parity with the final Docker deployment.

Advanced Architectural Concepts (For MVP/Production)

- **1. Model-as-a-Service (MaaS) Abstraction:** Abstract the LLM inference behind a clear **interface**, allowing for the "Model Cascade" to become a runtime feature (e.g., serving a small, fast model for standard tasks and falling back to a large, accurate MoE model for complex/long-context requests).
- **2. FinOps-Aware Microservices:** Implement **cost-aware scheduling** (e.g., using Kubernetes HPA with custom metrics like GPU utilization or request backlog) and consider **Model Offloading** (CPU fallback for low-priority tenants) to optimize cloud/hardware costs, which is critical for profitability.
- **3. Idempotent Ingestion Pipeline:** Design the document ingestion and embedding pipeline to be **idempotent**. This means repeated processing of the same document must yield the same result and not introduce duplicates or inconsistencies in the **Qdrant** vector store, crucial for data integrity in a multi-tenant system.

6. Common Technical Pitfalls & Mitigation (PoC Risks)

Pitfall	Probability	Mitigation Strategy
Model Environment Dependencies	High	Use a single base Docker image (e.g., an official PyTorch/CUDA image) for both fine-tuning (Days 1-3) and inference (Days 7-9). Utilize Poetry/Pipenv for strict dependency locking.
Load Test Resource Starvation	Medium	Execute load tests outside of the main PoC development environment. If using K3s/Docker Compose, allocate explicit, generous resource limits (CPU/GPU memory) to the inference container to avoid false-negative performance results.
Data Quality/Annotation Bottleneck	High	Scope the training corpus aggressively (e.g., \le 1000 documents). Focus on active learning or a smaller, high-fidelity human-validated subset (e.g., 200 documents) to ensure the 95% accuracy KPI is measured against a reliable \text{ground truth}.
Tenancy Data Leakage	Medium	Implement the Tenant ID as the primary partitioning key/metadata filter in Qdrant .

Pitfall	Probability	Mitigation Strategy
		Perform a dedicated security check (Day 13) to ensure no vector search query returns results from an unauthorized tenant ID.